



Ontology Foundry Efforts

Dr. John Beverley

President, *National Center for Ontological Research, Inc.*

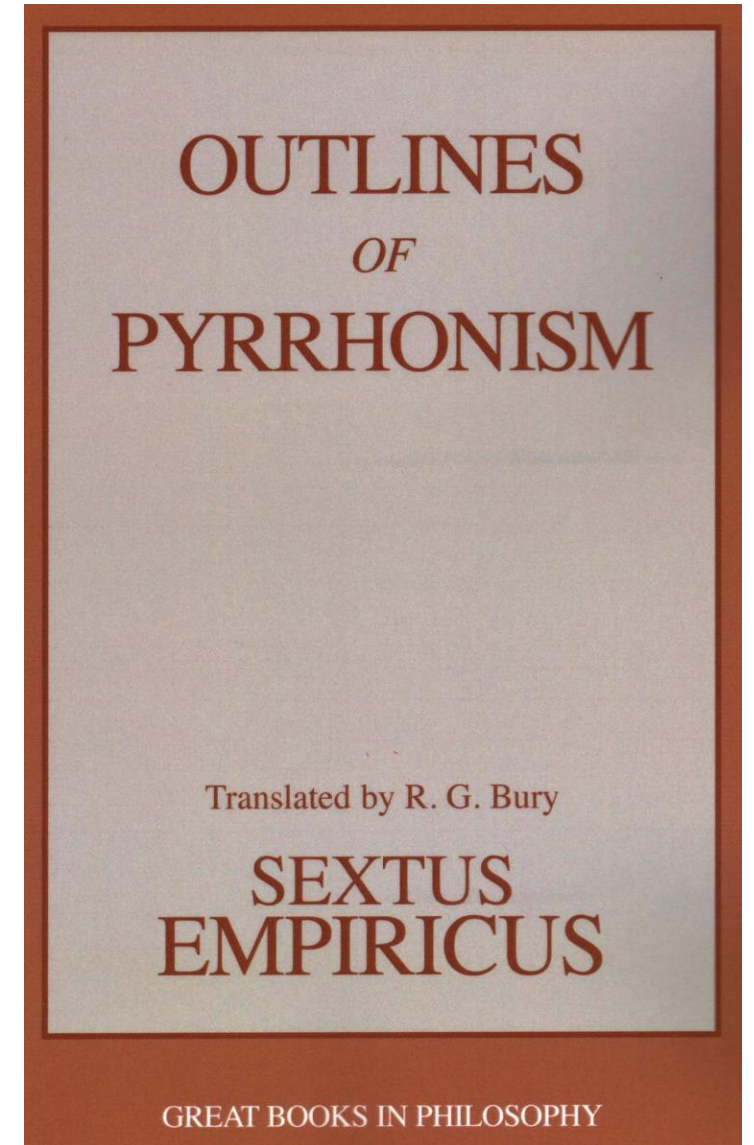
Assistant Professor, *University at Buffalo*

Associate Director, *Institute of AI and Data Science*

CEO, *Acacia Knowledge Systems Inc.*

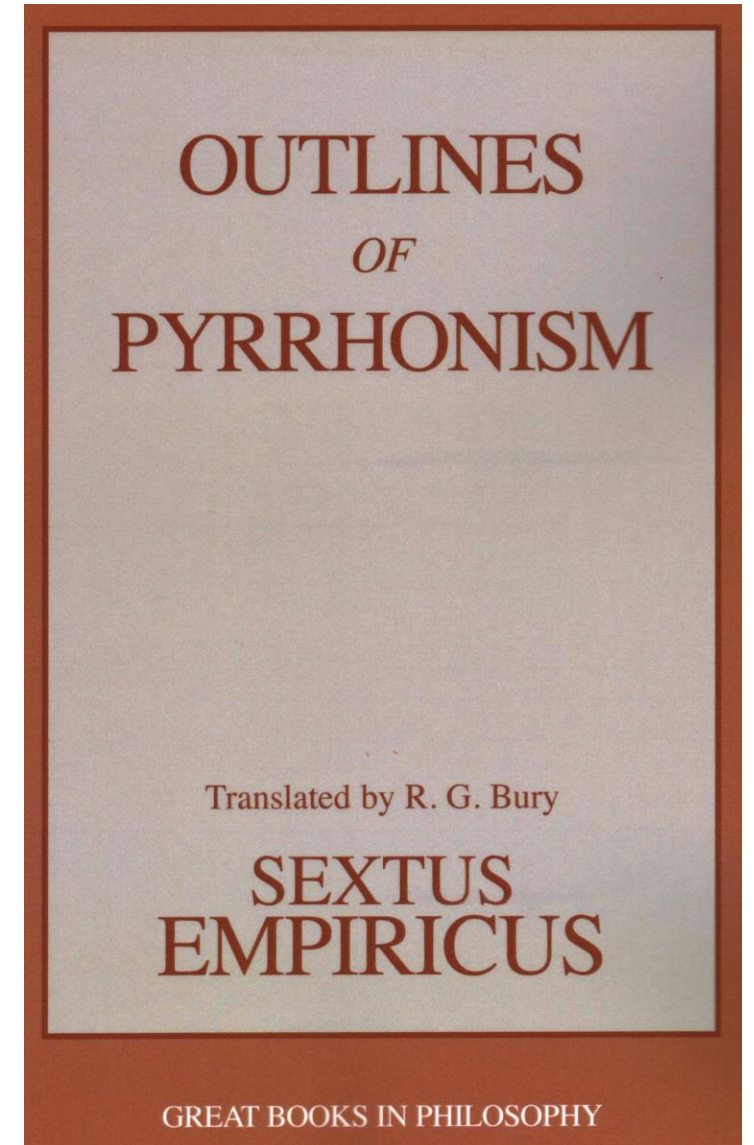
Pyrrhonism

- An ancient skeptical philosophy that taught suspension of belief was the only way to maintain an inquisitive attitude
- They were known for giving arguments on either side of claim, in the interest of showing there was no real truth
- They were also known for giving **bad arguments** on occasion



Pyrrhonism

- In his defense of Pyrronism, Sextus Empiricus wrote on this point:
- Physicians do not always give the strongest medicine, for the patient's body is not ready; they instead give weaker doses, building a patient's constitution, so they are **ready for the strongest medicine**



Building Knowledge Models Can be Easy

Building a Knowledge Graph From Scratch Using LLMs

Turn your Pandas data frame into a knowledge graph using LLMs. Build your own LLM graph-builder from scratch, implement LLMGraphTransformer by LangChain, and QA your KG.



Cristian Leo

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How to Build a Knowledge Graph in Minutes (And Make It Enterprise-Ready)

I tried and failed creating one—but it was when LLMs were not a thing!



Thuwarakesh Murallie

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Constructing knowledge graphs from text using OpenAI functions

Seamlessly implement information extraction pipeline with LangChain and Neo4j



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Building Knowledge Models Can be Easy

AWS Database Blog

Build a knowledge graph in Amazon Neptune using Data Lens

by Russell Waterson | on 05 MAY 2021 | in [Amazon Neptune](#) | [Permalink](#) | [Comments](#) | [Share](#)

This is a guest post by Russell Waterson, Knowledge Graph Engineer at Data Lens Ltd.

Customers use knowledge graphs to consolidate and integrate information assets a

Building [knowledge graphs](#) by getting data from disparate existing data sources can be complex. Project planning, project management, engineering, maintenance and release complexity and time to build a platform to populate your knowledge graph database

The [Data Lens](#) Ltd. team have been implementing Knowledge Graph solutions as a new now released tooling to greatly reduce the time and effort required to build a Know

Use [Data Lens](#) to make building a knowledge graph in Amazon Neptune faster and simpler configuration.

How to build a knowledge graph in 7 steps



[John Stegeman](#), [Kara Doriani O'Shee](#)

March 12, 2025 14 min read

SILOS

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**Building ontologies or knowledge graphs
according to standards can be HARD**

But that's how we avoid silos...

Disciplined Semantics to Governance

- Ontology methodology clarifies what kinds of entities and relations we are talking about
- BFO supplies top-level discipline but once many communities build together, the challenge becomes coordination, validation, and reuse
- And identifying **infrastructure that lets ontologies function as a community asset**



Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

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How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

It Takes a Village

- The same label can designate different kinds of things in the same community, e.g. **solitude**, **target**, **biomarker**
- The same format can carry different assumptions about identity, time, evidence, and context
- Data can be exported while the meaning needed to interpret it remains inside a platform, workflow, or codebase

Smithery

- Coordinate **meaning**: identifiers, definitions, relations, and axioms
- Coordinate **method**: modeling principles, reuse expectations, and quality criteria
- Coordinate **lifecycle**: review, validation, release, deprecation, and maintenance
- Coordinate **community**: domain experts, ontology engineers, users, and governance



Open Biological and Biomedical Ontologies

- In 2005, a consortium of biologists decided to create standards for ontology development
- Such as requiring ontologies be open-source, have documentation, include definitions for vocabulary terms...



Overview

Open (principle 1)

Common Format (principle 2)

URI/Identifier Space (principle 3)

Versioning (principle 4)

Scope (principle 5)

Textual Definitions (principle 6)

Relations (principle 7)

Documentation (principle 8)

Documented Plurality of Users
(principle 9)

Commitment To Collaboration
(principle 10)

Locus of Authority (principle 11)

Naming Conventions (principle 12)

Notification of Changes (principle 13)

Maintenance (principle 16)

Responsiveness (principle 20)

Industrial Ontologies Foundry

- In 2016, the Systems Integration Division at the National Institute of Standards and Technology (NIST) brought together industrial manufacturing researchers and ontologists.
- Who aimed to promote best practices for ontology development and preserve interoperability.



- 1 Introduction
- 2 IOF Ontology Architecture
- 3 Common Representations
- 4 Open Availability and Copyright
- 5 Scope and Context
- 6 Modularity
- 7 Ontological Considerations
- 8 Foundational Ontology – Use
- 9 Documentation
- 10 Textual Definitions
- 11 Identifiers and Naming Conventions
- 12 IRI and Identifier Space
- 13 Versioning
- 14 Maintenance
- 15 Contributed Ontologies: – Use
- 16 Collaboration

National Security Ontology Foundry

- 2016 also witnessed chartered efforts to promote interoperability among ontologies created within and across USG; formally recognized in 2023 by IC and DOD chief data officers
- Aimed at constructing vetted ontology artifacts aligned with best practices, tailored to specific USG missions



P1 – OPENNESS	
<i>Summary</i>	
<i>Purpose</i>	
<i>Recommendations and Requirements</i>	
<i>Examples</i>	
P2 – MODIFICATION	
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<i>Implementation</i>	
P4 – IRI/IDENTIFIER SPACE	
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Federation

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Ontology “Foundries”

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P7 – TEXTUAL DEFINITIONS

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P8 – RELATIONS

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Purpose.....

P9 – DOCUMENTATION

**National Security
Ontology Federation**

FY26 National Defense Authorization Act

SEC. 1504. DEPARTMENT OF DEFENSE DATA ONTOLOGY GOVERNANCE WORKING GROUP.

(a) ESTABLISHMENT.—

(1) IN GENERAL.—The Secretary of Defense shall establish a working group to develop and implement a common data ontology and governance structure across the Department of Defense.

(2) DESIGNATION.—The working group established under to paragraph (1) shall be known as the “Department of Defense Data Ontology Governance Working Group” (in this section the “Working Group”).

(3) USE OF EXISTING STRUCTURES.—

(A) IN GENERAL.—Notwithstanding paragraph (1), the Secretary of Defense may designate an existing forum, council, or organizational body to serve as the Working Group if such entity satisfies the requirements of subsections (b) and (c).

(B) RULE OF CONSTRUCTION.—For the purposes of this section, a forum, council, or organizational body designated under subparagraph (A) is deemed to be a working group established by the Secretary of Defense under paragraph (1).

(b) PURPOSE.—The purpose of the Working Group is to inform and to progress the Department of Defense’s foundational data ontology work by developing and implementing domain-specific data ontologies and governance structures across the Department of Defense to expand data interoperability, enhance information

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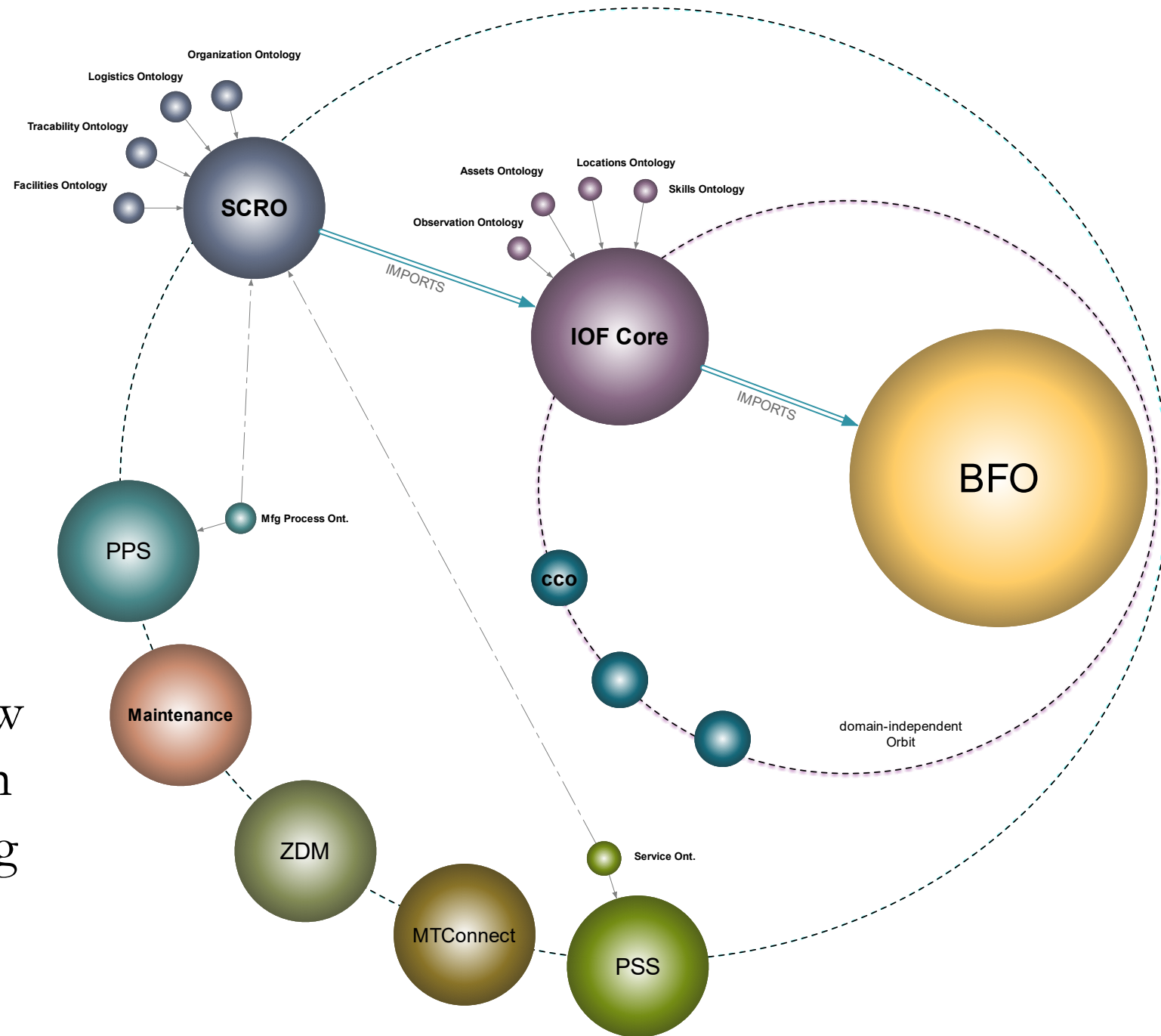
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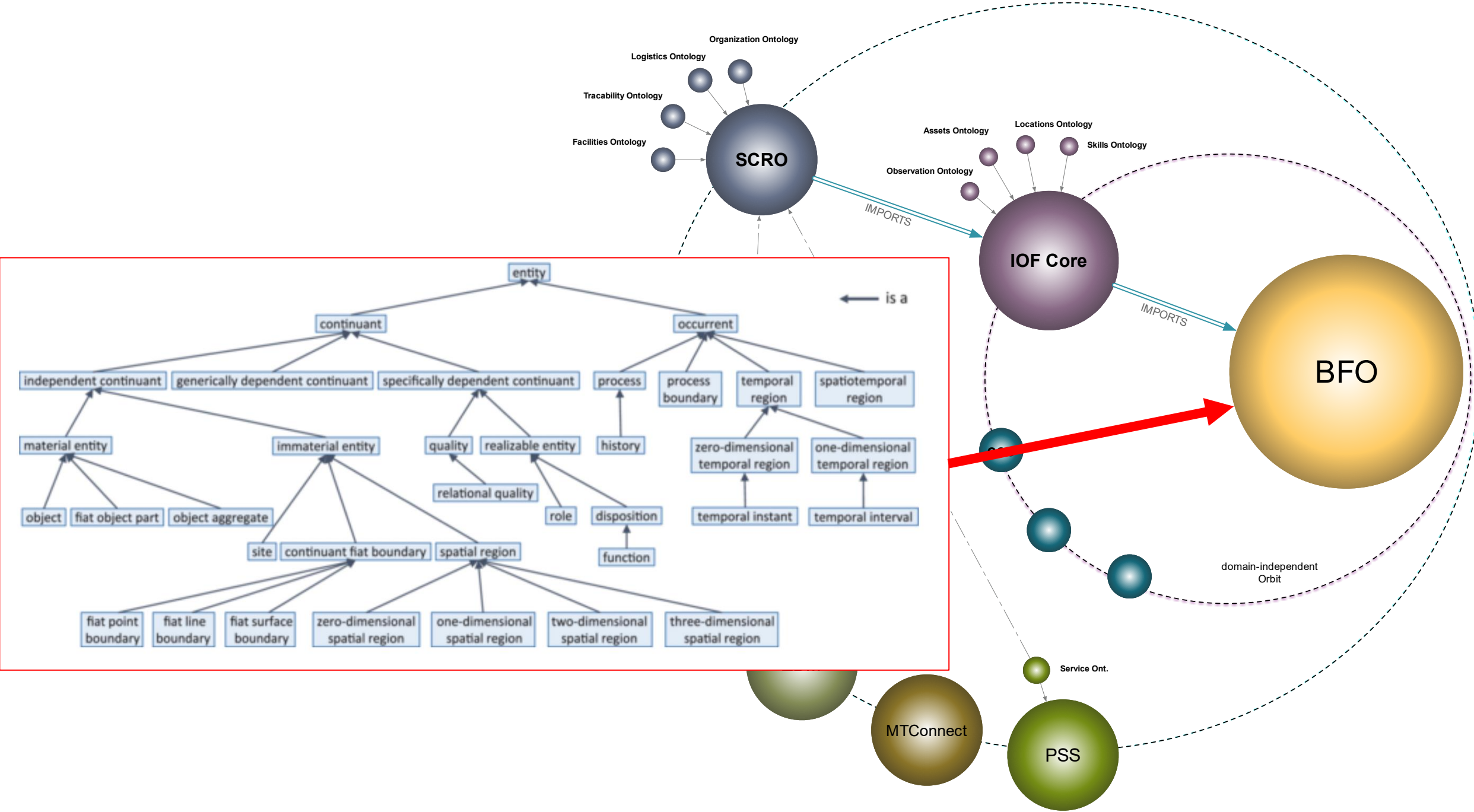
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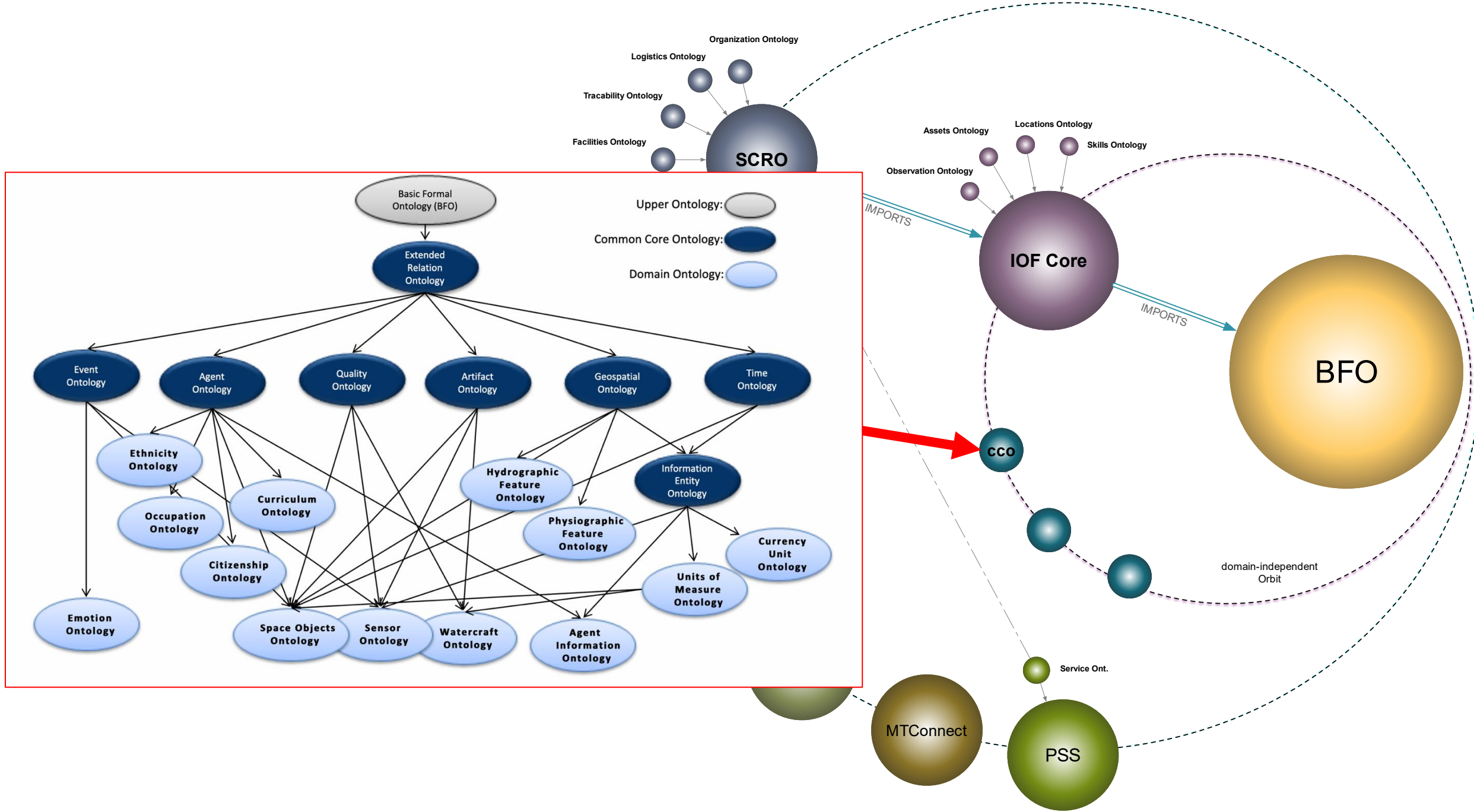
Enforcement

Ontologies in these Foundries are hubs logically extending from higher-level ontologies

Ontologies are extended by **downward population**, new classes have parent classes in a hierarchy ultimately leading to a BFO class







BFO Ecosystem

800+ Projects

BFO Basic Formal Ontology						
Home	GitHub	Guidebook	Publications	FOL	Users	Tutorials
Users						
Below you will find an alphabetical list of ontologies and institutions/groups using BFO						
Ontologies						
• ACGT Master Ontology (ACGT MO) • Actionable Intelligence Retrieval System (AIRS) • Addiction Ontology (Addict-O) • Additive Manufacturing Ontology (AMO) • Adolescent Depression Ontology (ADO) • Adverse Event Reporting Ontology (AERO) • AFO Foundational Ontology • African Wildlife Ontology (AWO) • Agronomic Linked Data (AgroLD) • Agronomy Ontology (AGRO) • Aircraft System Ontology • Algorithm-Implementation-Execution Ontology Design Pattern • Alzheimer Disease Ontology (ADO) • Alzheimer's Disease Diagnosis Ontology (ADDO) • Anatomy of the Insect Skeleto-Muscular system ontology (AISM) • Animals in Context Ontology (ACO) • Anthropological Notation Ontology (ANNO) • Antimicrobial-Microorganism Ontology • Apollo Structured Vocabulary (Apollo-SV) • Argument Ontology (ARGO) • ARIES (Arkansas Imaging Enterprise System) Knowledge Graph • Asset Management Ontology (AMODO) • Autism-DSM-ADI-R Ontology (ADAR) • Bacterial Clinical Infectious Diseases Ontology (BCIDO) • Baden Württemberg Materials Digital Domain Ontology (BWMD) • Bank Ontology • Battle Management Ontology (BMO) • Behavior Change Intervention Ontology (BCIO) • Behaviour Change Technique Ontology • Behavior Perspective Model (BPM) • Beta Cell Genomics Application Ontology (BCGO) • Bio-Knowledge Network Ontology (BioKNO) • BioAssay Ontology • Bioinformatics Web Service Ontology (OBIWS) • Biological Collections Ontology (BCO) • Biomedical Ethics Ontology • Biomedical Grid Terminology (BiomedGT, retired) • Biomedical Study - Lifecycle Management (BMS-LM) core ontology • Biomimetic Ontology • BioTop: a biomedical top-domain ontology						

• OntoAlign++: A Combined Strategy for Improving Ontologies Alignment • OntoBuildableSpace Ontology • OntoDM Core • OntoForInfoScience • Ontologies for Representing Surgical Procedure Models (OntoSPM) • Ontologized Minimum Information About Biobank data Sharing (OMIAS) • Ontology Based Clinical Decision Support System for Geriatrics • Ontology Based Decision Support System for Tuberculosis Management • Ontology for Adverse Events (OAE) • Ontology for Autism Spectrum Disorder • Ontology for Biobanking (OBIB) • Ontology for Biofilms (BIFO) • Ontology for Biomedical Investigations (OBI) • Ontology for Cancer research variables (OCRV) • Ontology for Computable Eligibility Criteria - Hepatitis C Virus (OCEC-HCV) • Ontology for Dengue Fever (IDODEN) • Ontology for Documentation of Variable/Data Source Selection (ODVDS) • Ontology for Drug Discovery Investigations (DDI) • Ontology for Energy Investigations (OEI) • Ontology for Functionally Graded Materials (OFGM) • Ontology for General Medical Science (OGMS) • Ontology for Genes and Genomes - Mouse (OGG-MM) • Ontology for Genetic Interval (OGI) • Ontology for Guiding Appropriate Antibiotic Prescribing • Ontology for Information Science (OntoForInfoScience) • Ontology for Laparoscopic Surgeries (LapOntoSPM) • Ontology for MicroRNA Target Prediction (OMIT) (here) • Ontology for Newborn Screening and Translational Research (ONSTR) • Ontology for Next Generation Sequencing Experiments (NGS Ontology) • Ontology for Nutritional Epidemiology (ONE) • Ontology for Nutritional Studies (ONS) • Ontology for Pain and Related Disability, Mental Health and Quality of Life • Ontology for Parasite LifeCycle (OPL) • Ontology for Periodontitis (PERIO) • Ontology for Petroleum Production • Ontology for Prognostic Health Management (PHM) in Spacecraft Avionics • Ontology for Stem Cell Investigations (OSCI) • Ontology for the Documentation of Variable Selection and Data sources • Ontology for Thoracentesis • Ontology of Autonomous Driving Based on the SAE J3016 Standard • Ontology of Arthropod Circulatory Systems (OArCS) • Ontology of Biological and Clinical Statistics (OBOS) • Ontology of Cancer Related Social-Ecological Variables (OCRSEV) • Ontology of Card Sleights • Ontology of Cardiovascular Drug Adverse Events (OCVDAE) • Ontology of Chinese Medicine for Rheumatism (OCMR) • Ontology of Clinical Research (OCRe) • Ontology of Commercial Exchange (OCE) • Ontology of Data Mining (OntoDM)

• Shop-Floor Digital Twin (DT) ontology • Situated and Interactive Multimodal Conversations • Situation Awareness Ontology (SAO) • Sketch Map Ontology • Skin Physiology Ontology (SPO) • Sleep Domain Ontology (SDO) • SMART Protocols: SeMAnTic RepresenTation for Experimental Protocols • Smart Ultrasound in Obstetrics and Gynecology (SUOG) Ontology • SNOMED CT (SCT) Standard Ontology • Social Determinants of Health Ontology (SDoHO) • Social Psychology Ontology (SPO) • Socialog Ontology for Social Simulation • Software Ontology (SWO) • Software, Disabilities and Competences Ontology (SODIC) • Soil Food Web Ontology • Space Domain Ontologies (SDO) • Space Object Ontology (SOO) • Spatial Graph Adapter (SGA) Ontology Design Pattern • Spatial Relation Ontology • Spatiotemporal Ontology for the Administrative Units of Switzerland (SONADUS) • Special Nuclear Materials Detection Ontology (SNM-DO) • Statistics Ontology (STATO) • STATO-LMM Linear Mixed Model Ontology • Style of Delivery Ontology • Subcellular Anatomy Ontology (SAO) • Suggested Ontology for Pharmacogenomics (SO-Pharm) • Supply Chain Traceability Ontology • Surface Water Ontology (SWO) • Survey Ontology • Sustainable Development and Climate (SDC) Ontology • Symptomatic Treatment of Multiple Sclerosis Ontology (STMSO) • Taxonomy for Rehabilitation of Knee Conditions (TRAK) • The Common Rule Ontology (CRO) • The Trope Ontology • Time Event Ontology (TEO) • Toxic Process Ontology (TXPO) • Trade-Space Analysis Tool for Constellations (TAT-C) ontology • Traditional Chinese Drugs Ontology (TCDO) • Translational Medicine Ontology (TMO)

<https://bfo-ontology.github.io/users.html>

Guiding Questions

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What can be done to minimize downstream alignment efforts across diverging communities?

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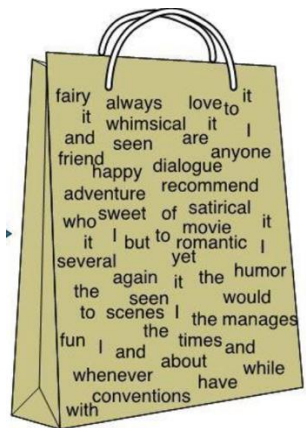
Policy, Funding, Inspired Volunteers, Technological Artifacts

What can be done to minimize downstream alignment efforts across diverging communities?

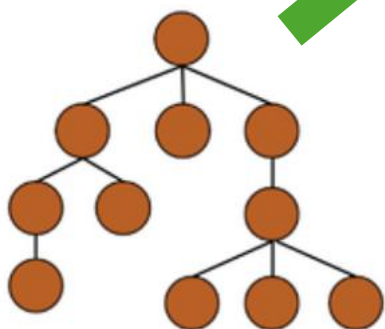
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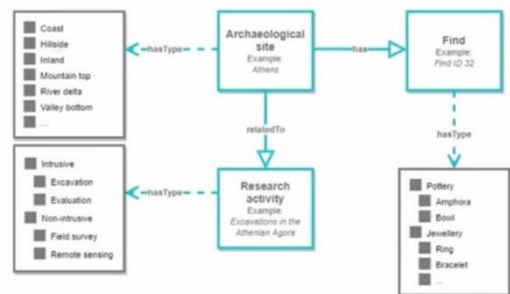
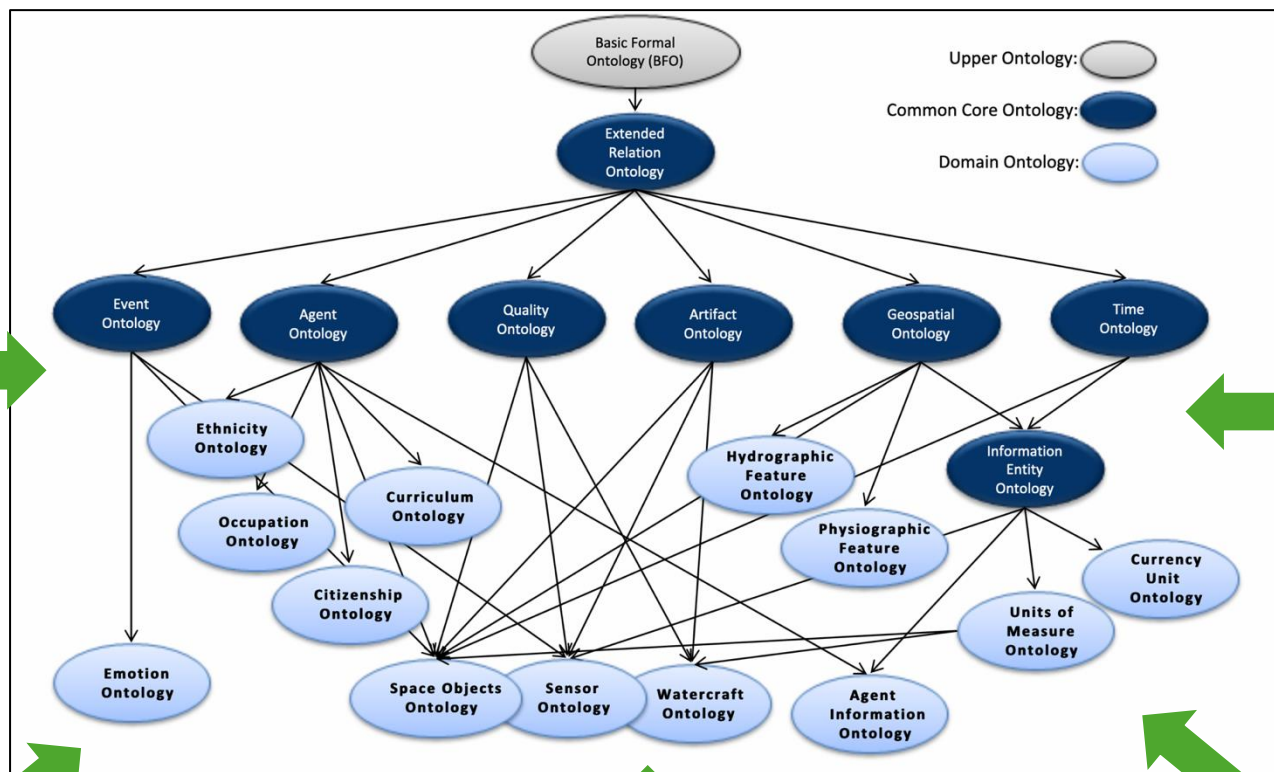
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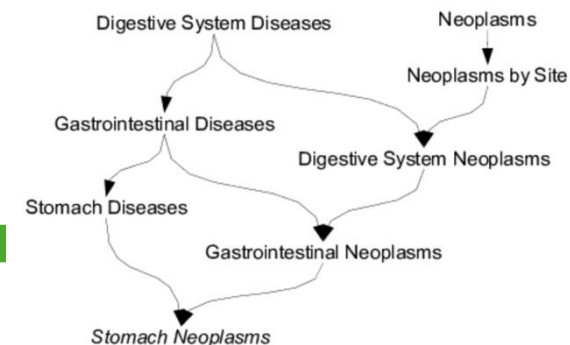
Bag of Words



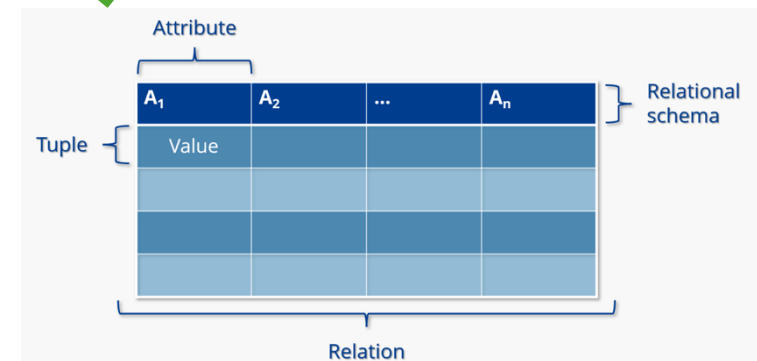
Ontology



Controlled Vocabulary



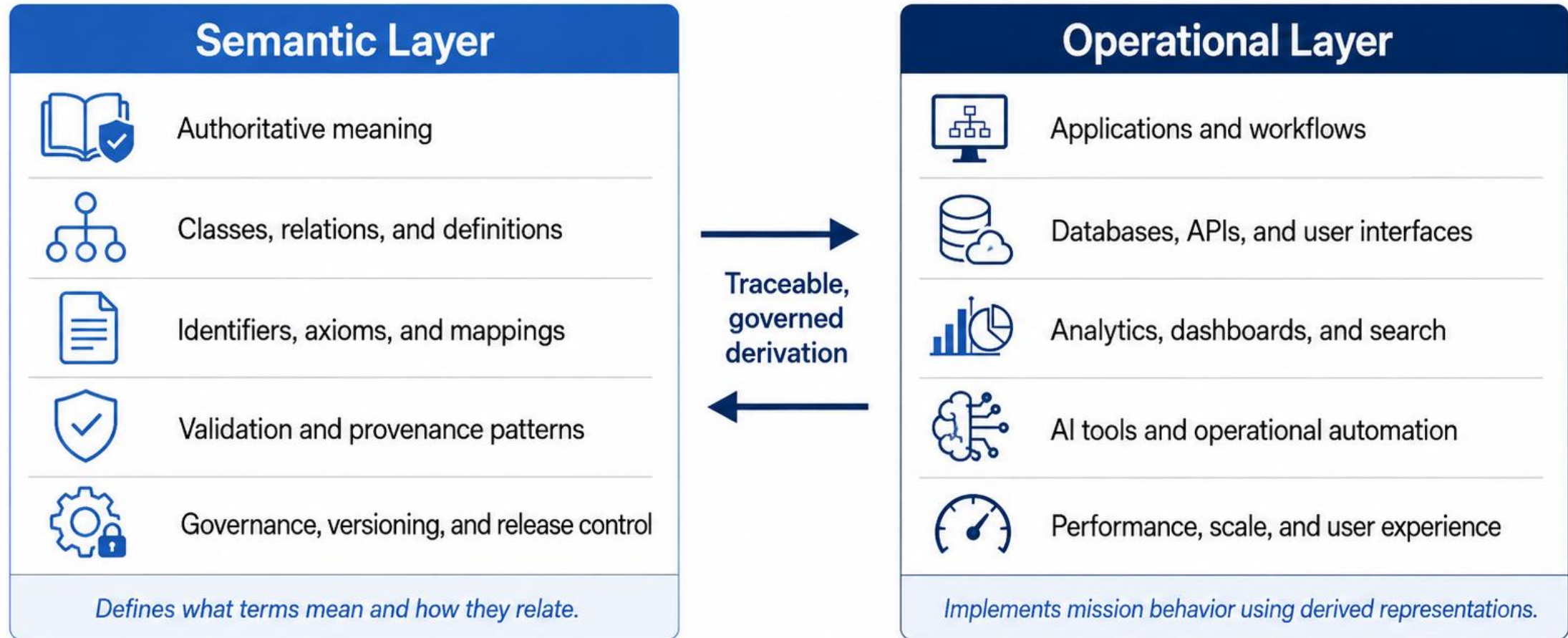
Thesaurus



Relational Database

Semantic Layer vs Operational Layer

Meaning is governed here; behavior is implemented there.



Meaning



Execution

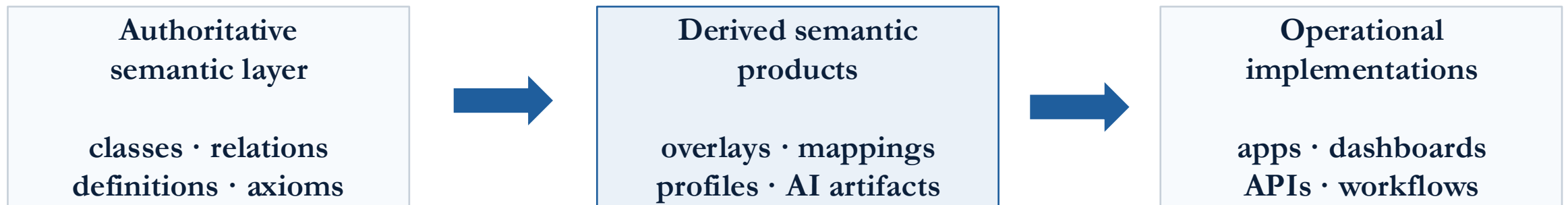
Semantic Layer

- An authoritative semantic layer facilitates traceable derived products
- Derived products should remain accountable to governed meaning
- Failure mode allows convenience artifacts to be ungoverned authorities



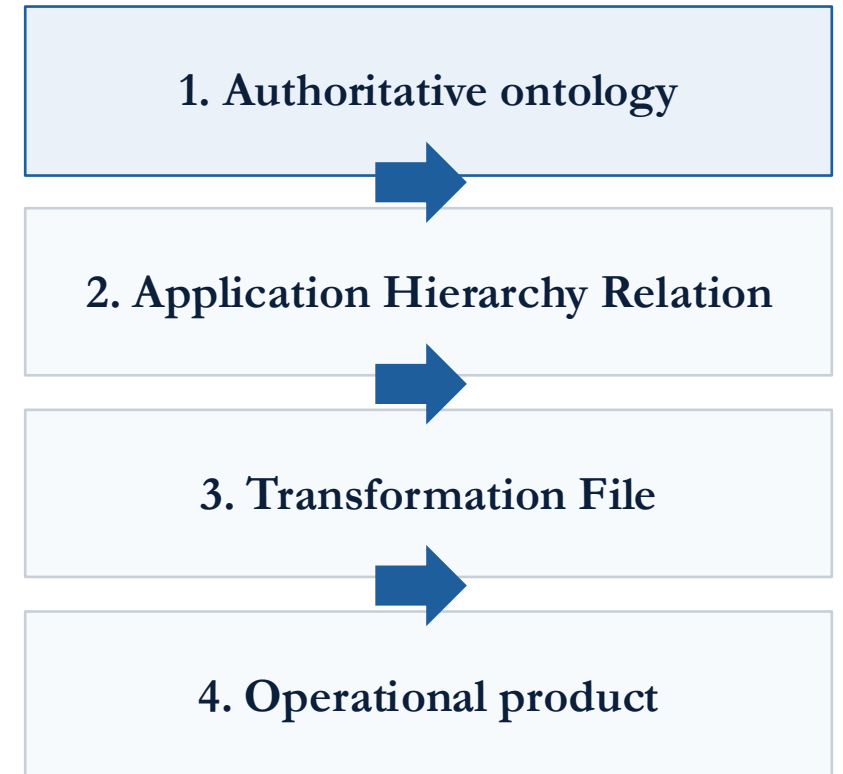
Derived Products

- Operational systems can use whatever runtime representations they need
- If they depend on governed meaning, they must remain traceable to it
- The governing question is whether meaning survives derivation



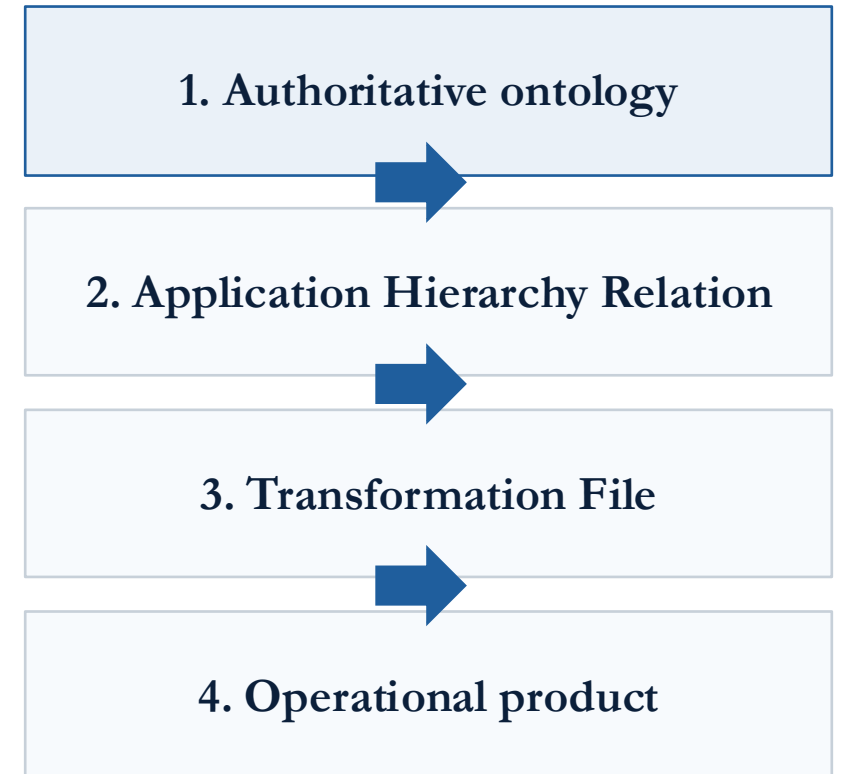
Derived Products

- Applications often need categories for search, navigation, workflow routing, etc.
- Some categories **may not** be in the authoritative semantic at all
- A governed relation can simulate hierarchy-like behavior so application gets the hierarchy it needs and the reference ontology remains clean

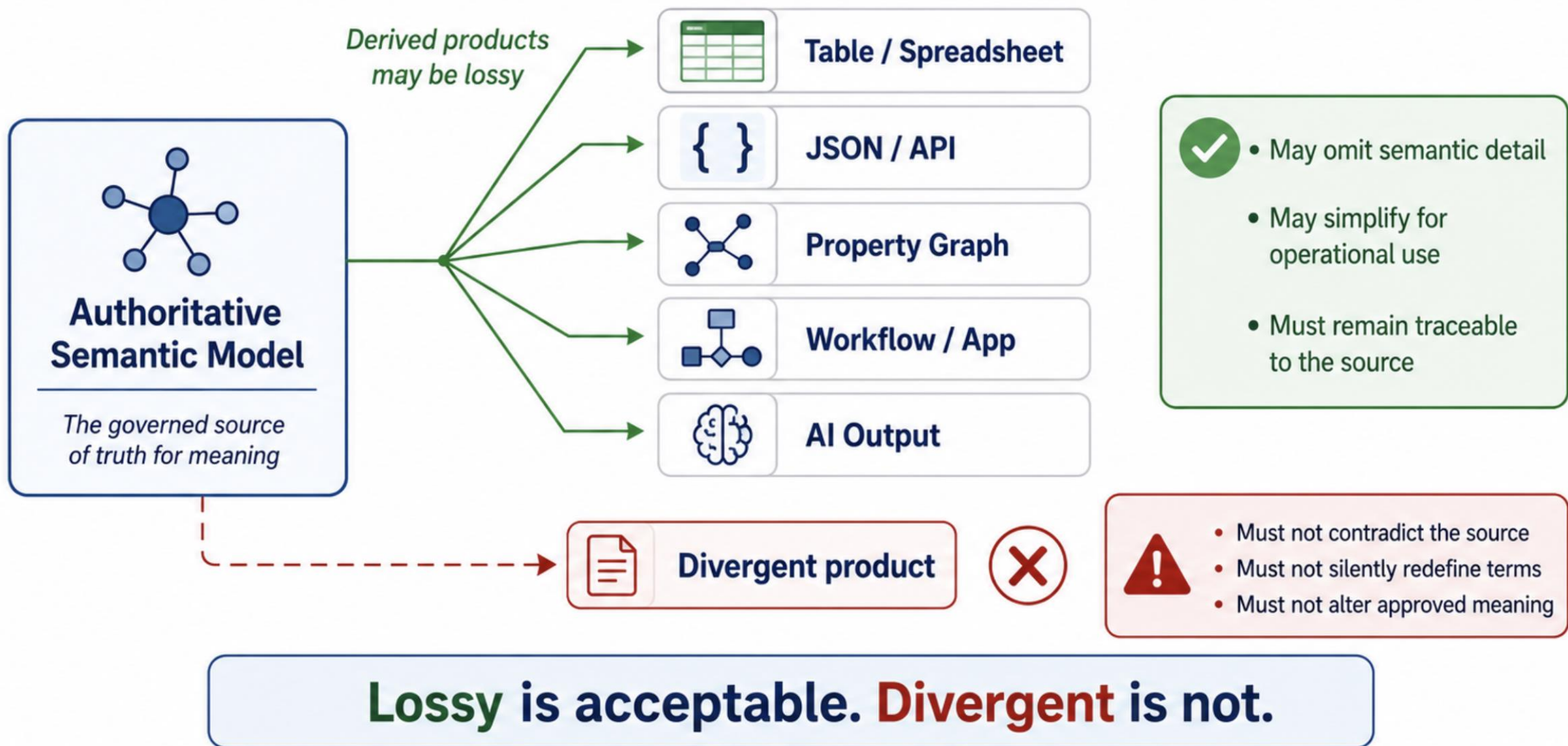


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Spoiler: John is going to teach you the logic you need to do this later



Summary

- We need governance around ontology development and maintenance to make good on the promises of ontology engineering
- Foundry efforts pursue this by aligning on core vocabulary and development principles
- While recognizing that minimizing downstream impact of misalignment requires governance around derived products used for specific applications

Bridge to Tradecraft

How do we build a worthwhile authoritative semantic layer?

- **Tradecraft** produces ontology content worth coordinating: scoped, justified, testable, and reusable
- By leveraging ontological insights in combination with semantic web technologies, as will see in later lectures

References

Derived Products from an Authoritative Semantics Layer

<https://ncor-network.org/blog/derived-products-from-authoritative-semantics>

Lossy, Never Divergent: Why Meaning Needs its Own Architecture

<https://ncor-network.org/blog/lossy-never-divergent-semantic-architecture>

The Round-Trip Test Every Data Platform Should Pass

<https://ncor-network.org/blog/round-trip-test-semantic-accountability>